



# MD. SAJJAD HOSSAIN

AVIONICS ENGINEERING STUDENT

EMBEDDED SYSTEMS / FLIGHT INTERFACES / SOFTWARE

Lalmonirhat, Bangladesh | sajjad00288@gmail.com

## PUBLIC LINKS

### GITHUB

github.com/Sajjadhossain9

### LINKEDIN

sajjad-hossain-369929299

### PORTFOLIO

sajjad-hossain0.netlify.app

## TECHNICAL SKILLS

- Embedded C and Python
- Arduino and ESP32
- Sensor interfacing
- Circuit integration
- PCB design and printing
- Micro-soldering
- Ground-control UI
- Responsive web development
- Data analysis and reporting
- Project coordination

## EDUCATION

B.Sc. in Avionics Engineering  
2023 - Present

Higher Secondary Certificate  
2022 - GPA 5.00

Secondary School Certificate  
2020 - GPA 5.00

## LANGUAGES

Bengali: Native  
English: Fluent in reading, writing and speaking

## PROFILE

Motivated avionics engineering student focused on electronics, programming, circuit design and embedded systems. I build across the complete signal path - from sensor interfacing and controller integration to dependable operator interfaces.

Hands-on university projects, research and software work have strengthened my problem-solving, teamwork and technical delivery. I am eager to keep learning and contribute to real aerospace and engineering systems.

## CORE FOCUS

01

### Embedded systems

Controllers, sensing, calibration and firmware.

02

### Flight interfaces

GCS concepts, telemetry and operator interaction.

03

### Circuits and PCB

Integration, fabrication, soldering and testing.

04

### Software systems

Web applications, UI design and data reporting.

## FEATURED ENGINEERING WORK

M-01

### Rocket Avionics - Team Shunno

Navigation, controller integration, GCS interface and recovery deployment testing.

MCU / NAVIGATION / RECOVERY

M-02

### Engine Health Monitoring

ESP32 and Hall-effect sensing for a 20cc light-aircraft engine monitoring system.

ESP32 / SENSORS / CALIBRATION

M-03

### Frequency-Hopping Radar Research

Simulation and system design focused on resilient radar operation.

RADAR / RF / SIMULATION

# EXPERIENCE AND PROJECT LOG

MD. SAJJAD HOSSAIN / AVIONICS - EMBEDDED - SOFTWARE

## AVIONICS AND EMBEDDED SYSTEMS

### TEAM SHUNNO

#### Rocket Avionics Team Member

Controller integration, navigation development, GCS UI and parachute deployment testing.

### AAUB DRONE CLUB

#### Executive Member - Software

Control-interface and UI/UX concepts for an aerial-surveillance drone and its ground-control software.

### DESIGN-BUILD-FLY TEAM

#### Avionics and Propulsion Team Member

Design, integration and testing support for avionics and propulsion systems on a student aircraft.

### AIRCRAFT SYSTEMS PROJECT

#### Arduino-Based RC Plane Project

Designed, programmed and integrated an Arduino-based remote-controlled aircraft.

### EMBEDDED MONITORING PROJECT

#### 20cc Engine Predictive Maintenance

ESP32 and Hall-effect sensing, circuit integration, calibration, firmware and monitoring UI.

## RESEARCH AND COMPETITION

### UNDERGRADUATE RESEARCH

#### Radar System and Frequency Hopping

Radar-system simulation focused on frequency-hopping techniques, resilient operation and implementation.

### TEAM AVYBARQ

#### Team Lead - IEEE CASS Student Design Competition

Coordinating circuit design, system development and technical delivery across a multidisciplinary team.

## LEADERSHIP, COMMUNITY AND DISCIPLINE

### Volunteer and Organizer

Red Crescent Youth, Volunteer for Bangladesh and Badhon Blood Donation Camp - humanitarian work, coordination, fundraising and event operations.

### Community Leadership

Coordinator at Sopnoshiri and Shekor social organizations; former president of Phulbary Science Club.

### Shito-ryu Karate - Brown Belt

Training at Freedom Martial Art Academy, developing discipline, physical conditioning and perseverance.

### Software Service Work

Part of a small technology team designing and maintaining websites, web applications and user interfaces.